



Visual Basic

Introduction to Visual Basic

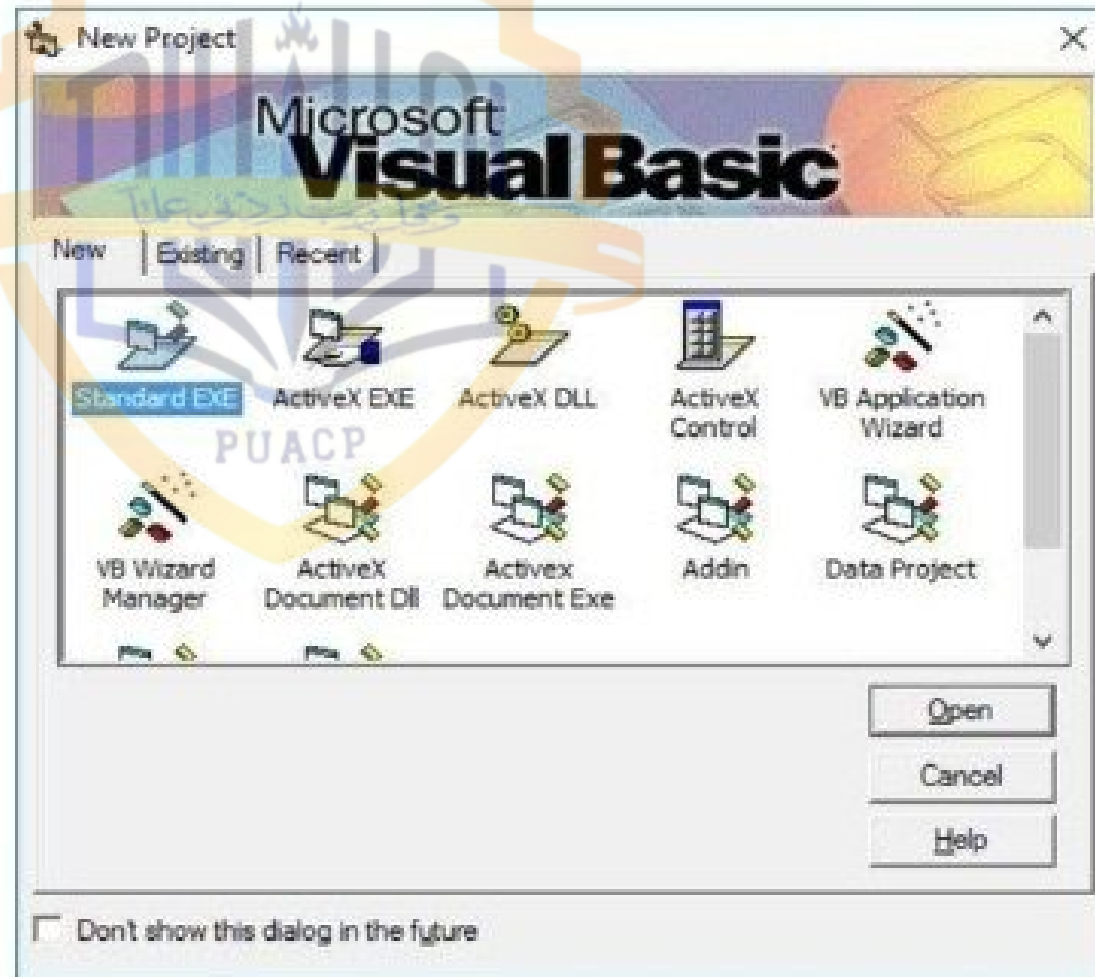
- VISUAL BASIC is a high level programming language evolved from the earlier DOS version called BASIC.
- BASIC stands for **B**eginners' **A**ll-purpose **S**ymbolic **I**nstruction **C**ode
- The program codes in Visual Basic resemble the English language
- Different software companies produce different versions of BASIC, such as Microsoft QBASIC, QUICKBASIC, GWBASIC, and IBM BASICA and so on.
- VISUAL BASIC is a VISUAL and events driven Programming Language

Divergences from the old BASIC.

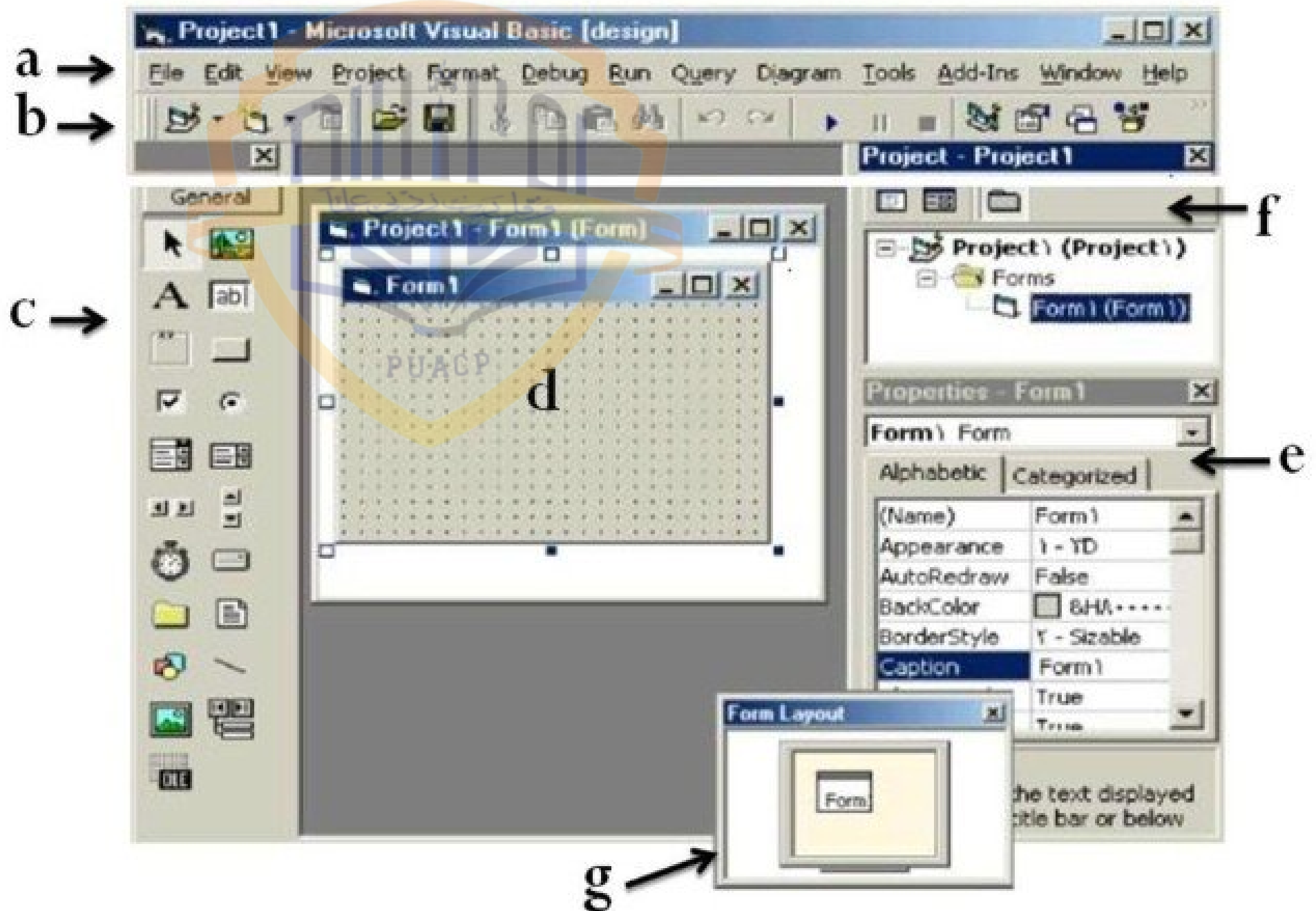
BASIC	VISUAL BASIC
Programming is done in a text-based environment and the program is executed sequentially.	Programming is done in a graphical environment.
Text-based procedure to design the interface	Design the interface by dragging and resizing the objects as well as changing their colors, just like any windows-based programs.

- Visual Basic is **event-driven** because users may click on a certain object randomly, so each object has to be programmed independently to be able to response to those actions (events).
- Examples of **events** are clicking a command button, entering text into a text box, selecting an item in a list box etc. Therefore, a VISUAL BASIC Program is made up of many subprograms; each with its own program code which can be executed independently and at the same time can be linked together in one way or another.

The Visual Basic Environment



The Visual Basic Environment



The Visual Basic Environment

a- Menu Bar: It contains a standard command like: File, Edit, View, Window, Help menus, and specific command such as: Project, Format, or Debug menus.

b- Toolbar: it contains the most commonly used commands (button), if clicked an action represented by that button is carried out.

c- Toolbox: it contains a collection of tools that are needed for project design.

d- form Designer: it is a window for each form to customize the designed interface of the application. Using the form designer, the user can add controls, graphics, and text to create the desired form appearance.

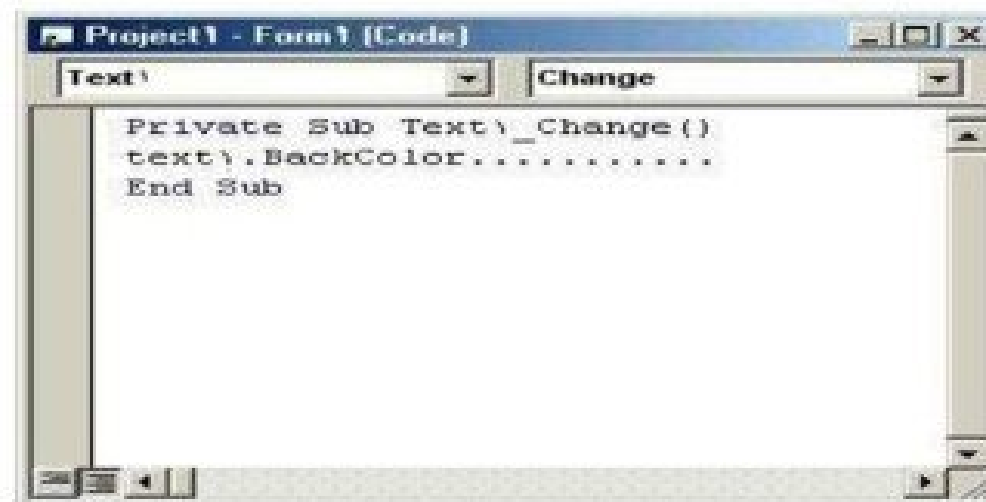
e- Properties Window: it is a List of properties settings for a selected form or a control. These properties are characteristics (such as size, visible, or color) of the selected object it provides an easy way to set properties.

The Visual Basic Environment

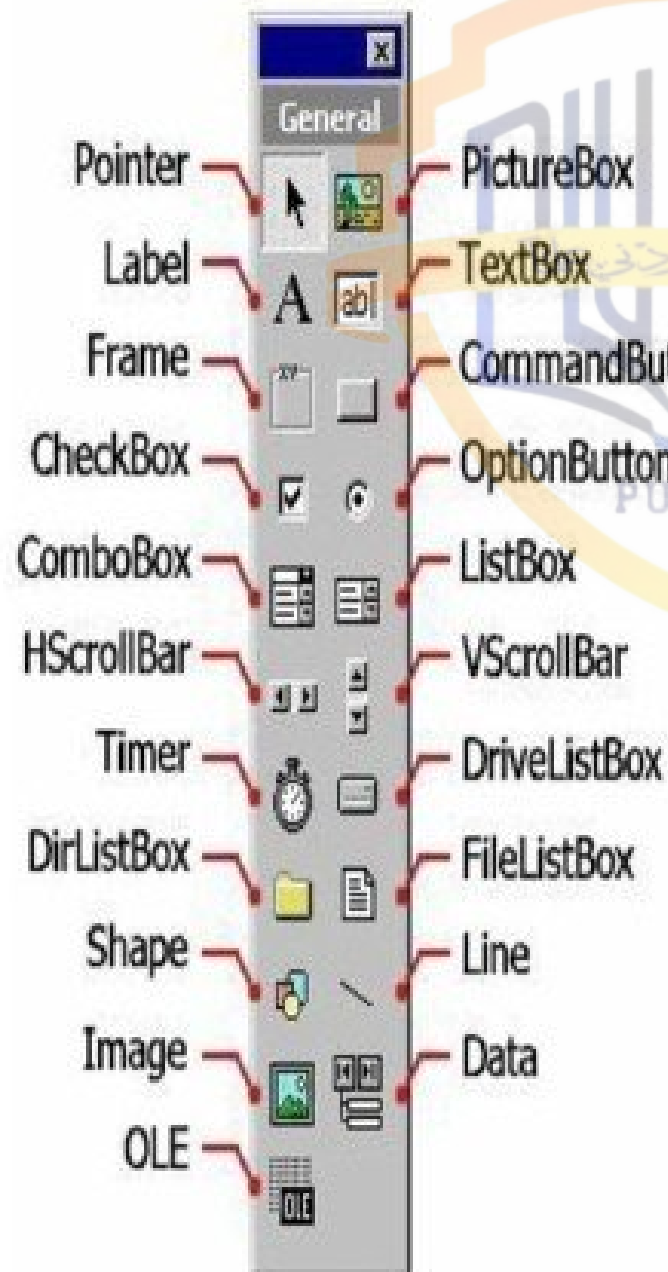
f- Project Explorer Window: it is a list of the forms and modules for the current projects. It is a hierarchical tree- branch structure, where the project at top of tree and other parts like forms ,modules) descend from this tree.

g- Form Layout Window: The Form Layout window is a small screen. Which is used to reposition the form of the application so that it appears in proper place when project is run.

h- Code Editor Window: Code Editor Window is used to write a VB code for an application. For each form there is a separate code editor window. It is displayed when user clicks on form or object in form.



The controls available for a Standard.exe project



The toolbox includes many tools and in general they are:

1) Pointer (not control) : used to select tools already on form

2) Picture box : Used to display images in any of the following

formats: BMP, DIB (bitmap), ICO (icon), CUR (cursor), WMF, (metafile), EMF (enhanced metafile), GIF, and JPEG.

3) Label : Fixed text appears on form for remark.

4) Textbox : For text edit .Like note pad.

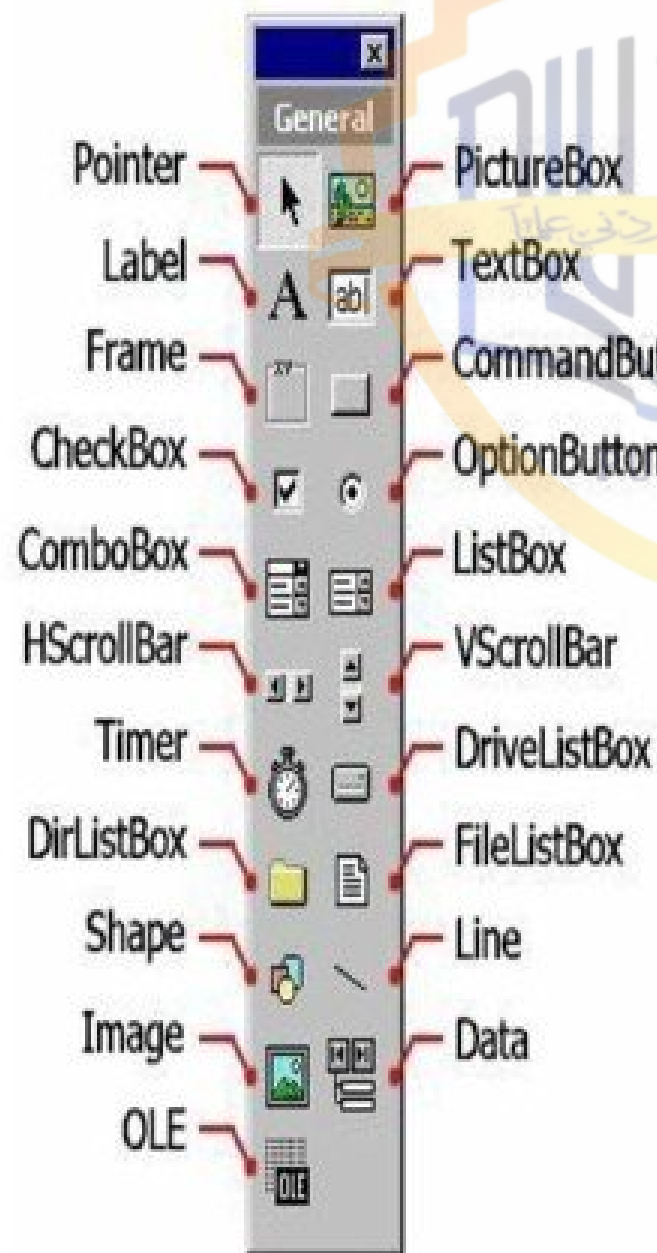
5) Frame : To group tools together (container).

6) Command button : Used as a switch (such as OK and Cancel) buttons. Code

is written in the Click event procedure of this control

7) Check box : For a yes/ no (true /false) selection.

The controls available for a Standard.exe project



8) Option button : For selection as group. Many options are placed inside container (grouped) (a Frame control). One control is selected from the group all others of the group are automatically deselected.

9) Combo box : consists of (list and arrow when clicked a small a list appears), if user selects item from the list, it will be displayed in TextBox. Vertical size is fixed.

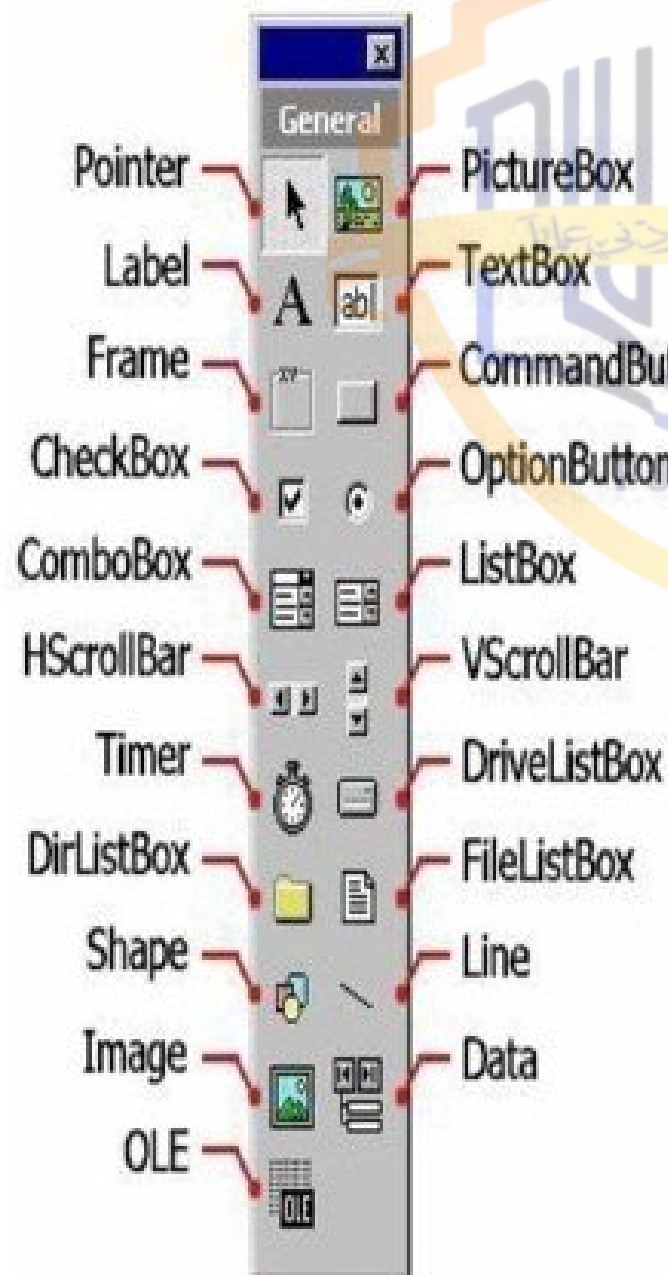
10) List box : For a list, user adds to and deletes from this list. It takes any size.

11) Horizontal Scrollbar : Create stand-alone Horizontal scroll bars.

12) Vertical Scrollbar : Create stand-alone vertical scroll bars.

13) Timer : Used to control object movement.

The controls available for a Standard.exe project



14) Drive List Box : It is a special ListBox filled automatically with names, of the files in a specified directory. It is a list invariant.

15) Dir List Box : It is a special ListBox filled with drives (Hard disc, Flopy, CD) in the system. It is an invariant.

16) File List Box : It is a special ListBox filled automatically with the names of all DirListBox. It is a list invariant.

17) Shape : Used only to display rectangles, circles, and ovals on the forms. Never raises any events

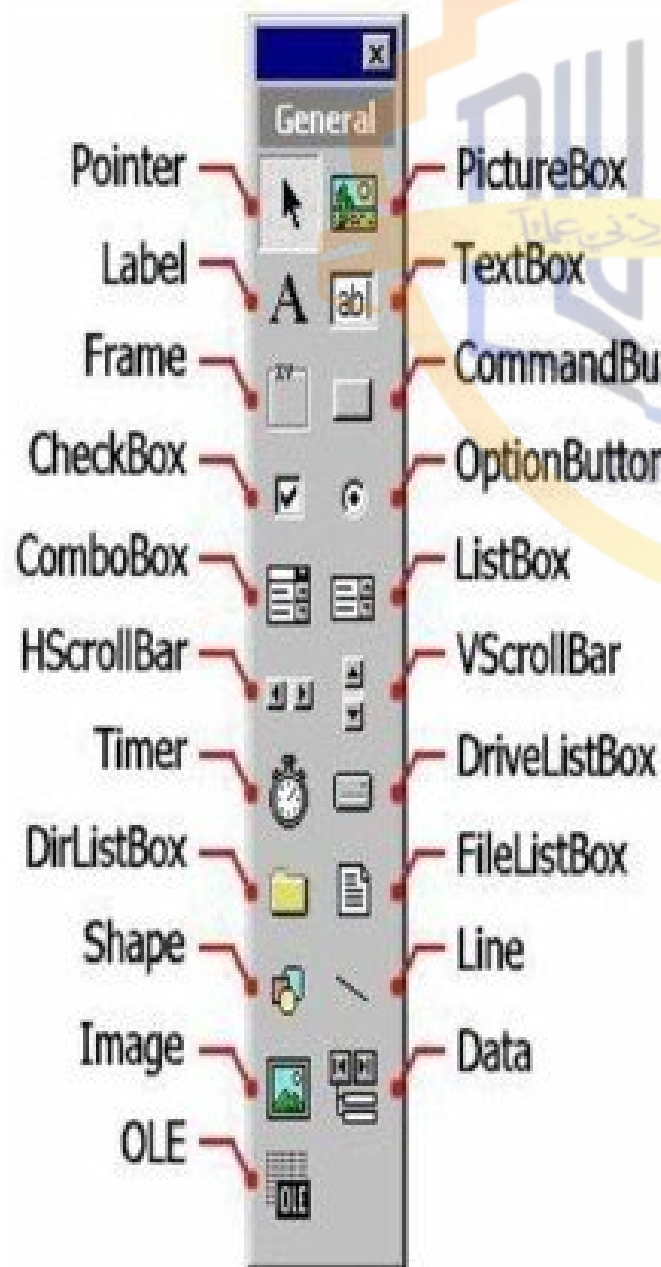
18) Line :Used only to display lines on the forms. It never raises any events.

19) Image : Used instead of PictureBox because it consumes fewer system resources.

20) Data : used for data base.

21) OLE : used for joining with another programs.

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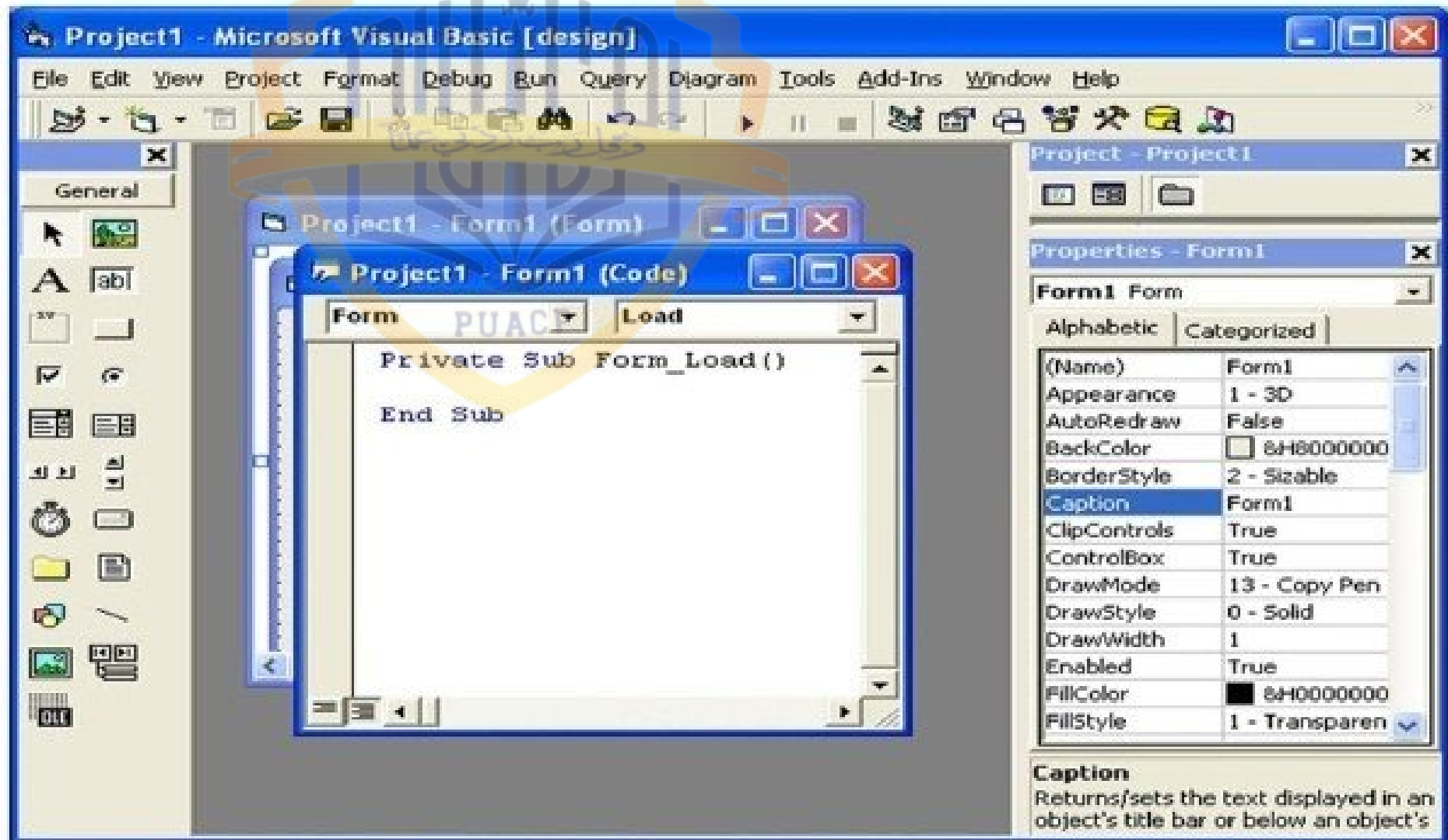
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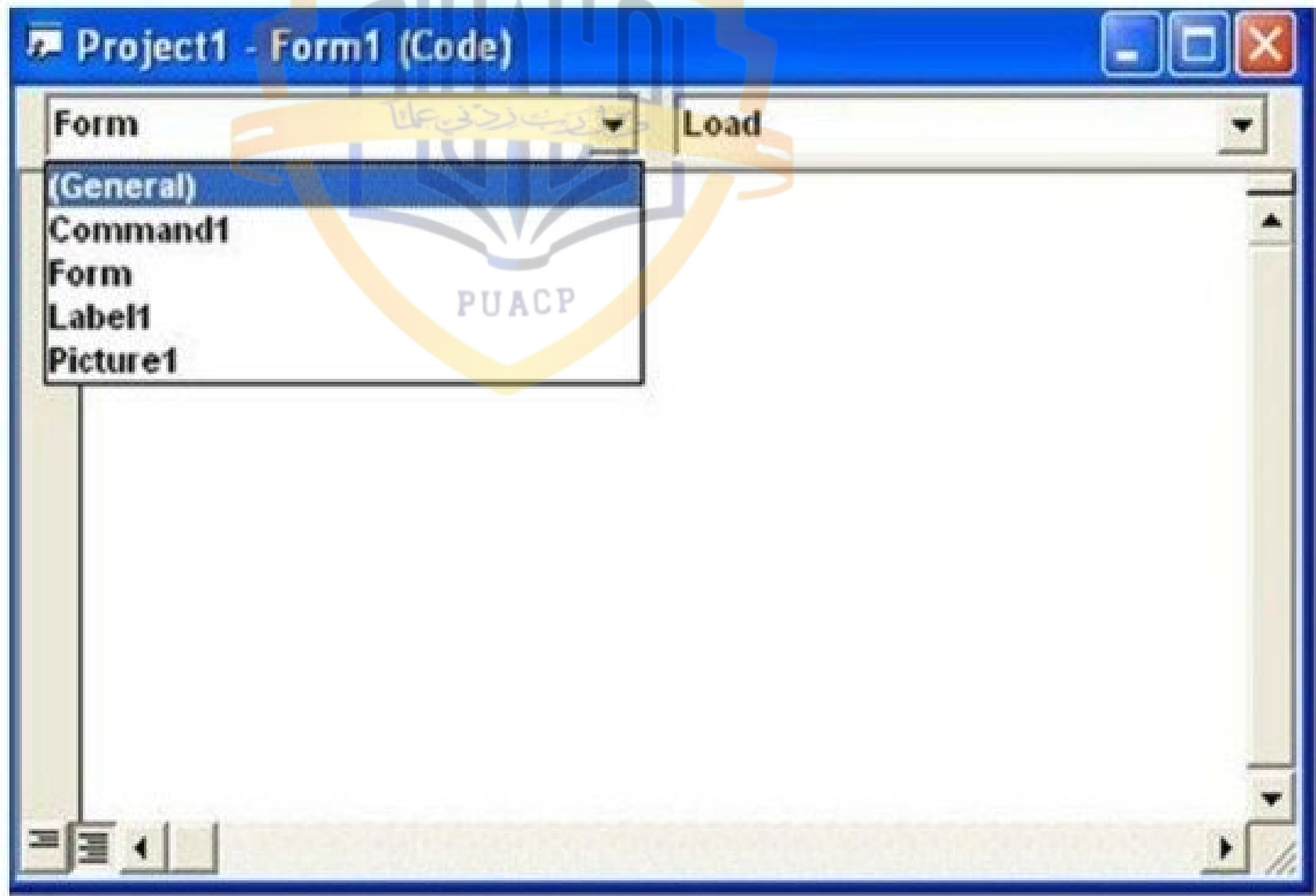
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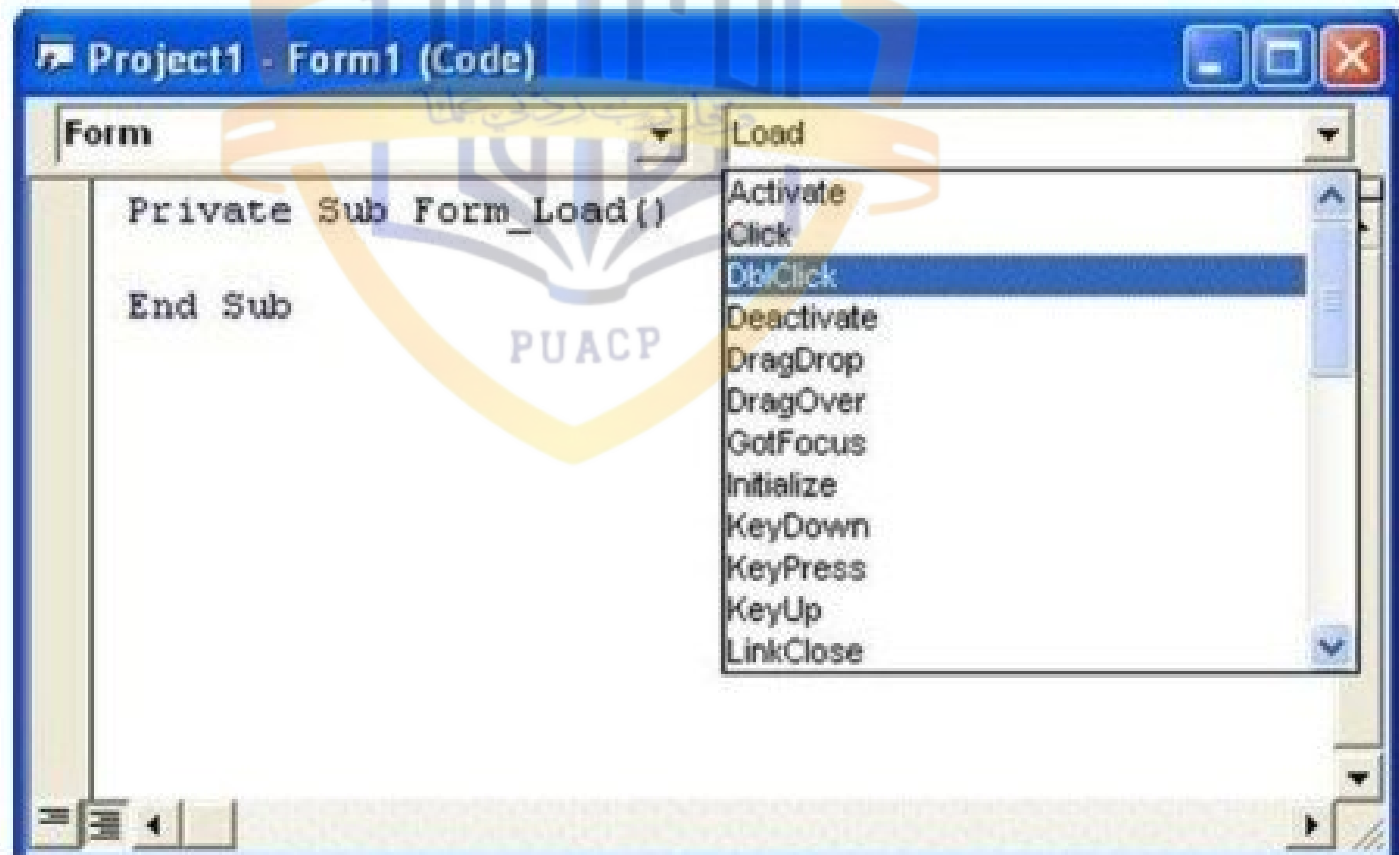
Introduction to Visual Basic



list of objects



list of procedures



Displaying the output

```
Private Sub Form_Activate ()
```

```
A = "Tom "
```

```
B = "likes "
```

```
C = "to "
```

```
D = "eat "
```

```
E = "burgers."
```

```
Print A & B & C & D & E
```

```
End Sub
```



+or the **&** operator to join two or more texts (string).

Summation of values using textbox, label, command button

```
Private Sub Command1_Click()  
'To add the values in TextBox1 and TextBox2  
    Sum = Val(Text1.Text) + Val(Text2.Text)  
'To display the answer on label 1  
    Label1.Caption = Sum  
End Sub
```

The screenshot shows a Windows application window titled "Form1". Inside the window, there are three text boxes and one command button. The first text box is labeled "Number 1" and contains the value "100". The second text box is labeled "Number 2" and contains the value "500". The third text box is labeled "Sum" and contains the value "600". To the right of the "Sum" text box is a command button labeled "Calculate".

Label	Value
Number 1	100
Number 2	500
Sum	600

Operators in Visual Basic

Arithmetical Operators

Operators	Description	Example	Result
+	Add	5+5	10
-	Subtract	10-5	5
/	Divide	25/5	5
\	Integer Division	20\3	6
*	Multiply	5*4	20
^	Exponent (power of)	3^3	27
Mod	Remainder of division	20 Mod 6	2
&	String concatenation	"George"&" "&"Bush"	"George Bush"

Operators in Visual Basic

Relational Operators

Operators	Description	Example	Result
>	Greater than	10>8	True
<	Less than	10<8	False
>=	Greater than or equal to	20>=10	True
<=	Less than or equal to	10<=20	True
<>	Not Equal to	5<>4	True
=	Equal to	5=7	False

Operators in Visual Basic

Logical Operators

Operators	Description	Example
AND	Compare two variables if a AND b equals	If x AND y ThenEnd If
OR	Compare two variables if a OR b equals	If x or y Then End If